

## SEMESTER S7

### ADVANCED COMPUTER NETWORKS

|                                      |                 |                    |                |
|--------------------------------------|-----------------|--------------------|----------------|
| <b>Course Code</b>                   | <b>PECST751</b> | <b>CIE Marks</b>   | 40             |
| <b>Teaching Hours/Week (L:T:P:R)</b> | 3:0:0:0         | <b>ESE Marks</b>   | 60             |
| <b>Credits</b>                       | 3               | <b>Exam Hours</b>  | 2 Hrs. 30 Min. |
| <b>Prerequisites (if any)</b>        | None            | <b>Course Type</b> | Theory         |

#### Course Objectives:

1. To give a comprehensive understanding of advanced networking concepts, including MPLS, VPNs, Data Center Networks, and Software-Defined Networking (SDN).
2. To impart the skills necessary to analyze, design, and evaluate complex networking architectures, addressing the challenges and emerging trends.

### SYLLABUS

| <b>Module No.</b> | <b>Syllabus Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>Contact Hours</b> |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <b>1</b>          | Review of Computer Networking Fundamentals - OSI and TCP/IP Models, Layers and Protocols, IP Addressing and Subnetting, Routing Protocols - RIP, OSPF, BGP;<br>QoS in IP networks - Random Early Detection, Protocols for QoS support - RSVP, RTP, Multiprotocol Label Switching (MPLS): Overview and Use Cases; Network Security Basics - Firewalls, ACLs, and NAT; Working of NAT; Virtual Private Networks (VPNs) - Types and Architectures; Overview of Data Center Networks: Key Components and Topologies; | <b>8</b>             |
| <b>2</b>          | DLL switching - Overview, VLANs, Inter-VLAN Routing; Spanning Tree Protocol (STP) - IEEE 802.1D, Rapid Spanning Tree Protocol (RSTP) - IEEE 802.1w, Multiple Spanning Tree Protocol (MSTP) - IEEE 802.1s, STP Enhancements - BPDU Guard, Root Guard, and Loop Guard;<br>Data Center Network Architectures - Traditional vs. Modern Data Center Designs (Spine-Leaf, Clos Networks), Ethernet Fabrics and TRILL;<br>Data Center Design Considerations - Scalability, Redundancy, and Latency.                     | <b>9</b>             |
| <b>3</b>          | SDN Architecture and Components - Control Plane, Data Plane, and Application Plane; OpenFlow Protocol and its Role in SDN; SDN Controllers - Ryu, OpenDaylight, and ONOS; SDN Use Cases - Traffic                                                                                                                                                                                                                                                                                                                | <b>9</b>             |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
|          | Engineering, Network Function Virtualization (NFV) - NFV Concepts, Virtualizing Network Functions and Services; NFV Infrastructure (NFVI) and Management (MANO); Service Function Chaining (SFC); NFV in Telecom Networks.                                                                                                                                                                                                                                                                                   |           |
| <b>4</b> | Data Center Interconnect (DCI) - Technologies for Data Center Interconnection(VPLS, OTV, and VXLAN), DCI Design and Deployment Considerations; Intent-Based Networking (IBN) - Introduction to Intent-Based Networking; Content Distribution on the Internet - Architectures for Information-Centric Networking; Content Naming, Routing and Caching, Security in Named Data Networking; Network Automation and Orchestration; Automation Tools - Ansible, Terraform; Orchestration Frameworks - Kubernetes. | <b>10</b> |

**Course Assessment Method**  
(CIE: 40 marks, ESE: 60 marks)

**Continuous Internal Evaluation Marks (CIE):**

| <b>Attendance</b> | <b>Assignment/<br/>Microproject</b> | <b>Internal<br/>Examination-1<br/>(Written)</b> | <b>Internal<br/>Examination- 2<br/>(Written)</b> | <b>Total</b> |
|-------------------|-------------------------------------|-------------------------------------------------|--------------------------------------------------|--------------|
| <b>5</b>          | <b>15</b>                           | <b>10</b>                                       | <b>10</b>                                        | <b>40</b>    |

**End Semester Examination Marks (ESE)**

*In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions*

| <b>Part A</b>                                                                                                                                                          | <b>Part B</b>                                                                                                                                                                                                                                                                         | <b>Total</b> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| <ul style="list-style-type: none"> <li>• 2 Questions from each module.</li> <li>• Total of 8 Questions, each carrying 3 marks</li> </ul> <p><b>(8x3 =24 marks)</b></p> | <ul style="list-style-type: none"> <li>• Each question carries 9 marks.</li> <li>• Two questions will be given from each module, out of which 1 question should be answered.</li> <li>• Each question can have a maximum of 3 subdivisions.</li> </ul> <p><b>(4x9 = 36 marks)</b></p> | <b>60</b>    |

### Course Outcomes (COs)

At the end of the course students should be able to:

| Course Outcome |                                                                                                                                                                           | Bloom's Knowledge Level (KL) |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>CO1</b>     | Explain and critically analyze advanced networking protocols and technologies, including MPLS, VPNs, and SDN, and their applications in modern networks                   | <b>K3</b>                    |
| <b>CO2</b>     | Demonstrate an understanding of data center network architectures, including the design considerations and protocols that ensure scalability, redundancy, and efficiency. | <b>K3</b>                    |
| <b>CO3</b>     | Use Software-Defined Networking (SDN) and Network Function Virtualization (NFV) to automate and optimize network operations.                                              | <b>K3</b>                    |
| <b>CO4</b>     | Explain emerging trends such as Intent-Based Networking (IBN) and network automation, applying this knowledge to modernize and innovate networking solutions.             | <b>K2</b>                    |

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

### CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

|            | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|------------|-----|-----|-----|-----|-----|-----|-----|-----|-----|------|------|------|
| <b>CO1</b> | 3   | 3   | 3   | 2   |     |     |     |     |     |      |      | 3    |
| <b>CO2</b> | 3   | 3   | 3   | 2   |     |     |     |     |     |      |      | 3    |
| <b>CO3</b> | 3   | 3   | 3   | 2   |     |     |     |     |     |      |      | 3    |
| <b>CO4</b> | 3   | 2   | 3   |     |     |     |     |     |     |      |      | 3    |

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

| Text Books |                                                                                                                                                        |                                                                    |                       |                  |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------|------------------|
| Sl. No     | Title of the Book                                                                                                                                      | Name of the Author/s                                               | Name of the Publisher | Edition and Year |
| 1          | Computer Networking: A Top-Down Approach                                                                                                               | James F. Kurose,<br>Keith W. Ross                                  | Pearson               | 8/e, 2022        |
| 2          | Data Center Virtualization Fundamentals: Understanding Techniques and Designs for Highly Efficient Data Centers with Cisco Nexus, UCS, MDS, and Beyond | Gustavo A. A. Santana                                              | CISCO Press           | 1/e, 2013        |
| 3          | MPLS and VPN Architectures                                                                                                                             | Jim Guichard, Ivan Pepelnjak, Jeff Apcar                           | CISCO Press           | 1/e, 2000        |
| 4          | High-speed networks and Internet: Performance and Quality of Service                                                                                   | William Stallings                                                  | Pearson               | 2/e, 2002        |
| 5          | Software Defined Networks: A Comprehensive Approach                                                                                                    | Paul Goransson,<br>Chuck Black,<br>Timothy Culver                  | Morgan Kaufman        | 2/e, 2016        |
| 6          | Information-Centric Networking (ICN): Content-Centric Networking (CCNx) and Named Data Networking (NDN) Terminology                                    | B. Wissingh, C. Wood, A. Afanasyev, L. Zhang, D. Oran, C. Tschudin | RFC 8793              | 2020             |

| Reference Books |                                                                  |                      |                       |                  |
|-----------------|------------------------------------------------------------------|----------------------|-----------------------|------------------|
| Sl. No          | Title of the Book                                                | Name of the Author/s | Name of the Publisher | Edition and Year |
| 1               | Cloud Networking: Understanding Cloud-based Data Centre Networks | Gary Lee             | Morgan Kaufman        | 1/e, 2014        |

| Video Links (NPTEL, SWAYAM...) |                                                                                                                             |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Module No.                     | Link ID                                                                                                                     |
| 1                              | <a href="https://archive.nptel.ac.in/courses/106/106/106106243/">https://archive.nptel.ac.in/courses/106/106/106106243/</a> |